Electricity

Mark scheme and worked solutions for 30 questions: 10 multiple choice, 20 structured.

1 · Part 1

Paper 1 & 2 Units [1]

Answer: B

Explanation

Charge is measured in coulombs, power in watts and resistance in ohms, so B is correct. A and D give the ampere, which is the unit of current, not charge. C and D give the joule, which is the unit of energy, not power. C also gives the volt, which is the unit of voltage, not resistance.

2 · Part 1

Paper 1 & 2 Mains electricity [1]

Answer: D

Explanation

A circuit breaker switches off the circuit when the current becomes too large, and it can then be reset, so D is correct. It does not increase the current (A); it is placed in the live wire, not the neutral wire (B); it is a fuse, not a circuit breaker, that must be replaced after it has operated (C).

3 · Part 1

Paper 1 & 2 Mains electricity [1]

Answer: A

Explanation

A cell or battery pushes charge in one direction only, so it supplies direct current; the mains supply keeps reversing direction, so it is alternating current. A is correct. B swaps the two, and C and D wrongly give both supplies the same type of current.

4 · Part 1

Paper 1 & 2 Energy and voltage in circuits [1]

Answer: C

Explanation

The resistance of an LDR decreases as the light on it gets brighter. With less resistance in the series circuit, the current increases, so the ammeter reading increases: C. A and B have the LDR resistance changing the wrong way. D forgets that a smaller resistance gives a larger current.

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5 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [1]

Answer: A

Explanation

In a solid metal such as copper, the current is a flow of negatively charged electrons, so A is correct. Protons (B) and positive ions (C) stay fixed in the metal and do not flow through the wire. Copper atoms (D) do not flow along the wire.

6 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [1]

Answer: D

Explanation

An LED lights up when there is a current in it, so it shows that a current is present: D. A fixed resistor (A), an LDR (B) and a thermistor (C) all allow a current, but none of them gives any visible sign that the current is there.

7 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [1]

Answer: B

Worked solution

At the junction, the current from the battery splits between the two branches, so it equals the sum of the branch currents: 0.30 + 0.20 = 0.50 A. Each branch is connected directly across the battery, so each lamp has the same 6.0 V across it.

8 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [1]

Answer: C

Worked solution

If the current were proportional to the voltage, doubling the voltage from 1.0 V to 2.0 V would double the current to 0.40 A. It only rises to 0.32 A, and each further 1.0 V adds less current than the one before. The graph would curve and flatten, which is the shape for a filament lamp.

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9 · Part 1

Paper 2 only | Electric charge | [1]

Answer: A

Explanation

Glass, polythene and rubber are all electrical insulators, so A is correct. B contains copper, C contains aluminium and D contains steel; all three are metals, which are electrical conductors.

10 · Part 1

Paper 2 only | Electric charge | [1]

Answer: C

Worked solution

Repulsion only happens between two objects with the same type of charge, so a repulsion is a reliable test of the type of charge. X is negative and repels Y, so Y is negative. Y repels Z, so Z is negative too.

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11 · Part 1 Paper 1 & 2 Mains electricity [6]

Mark scheme

(a) **[1]**

- power = current × voltage (1)

ALLOW P = I × V OR P = IV

ALLOW any correct rearrangement, e.g. current = power / voltage

(b) **[3]**

- converts the power: 1.8 kW = 1800 W (1)
- substitution into the rearranged equation: I = 1800 / 230 (1)
- 7.8 A (1)

ALLOW 7.83 A OR 7.826 A

ALLOW 0.0078 A for one mark only (power not converted to watts)

(c) **[2]**

- 13 A (1)
- it is the lowest fuse rating above the normal current of 7.8 A (1)

ALLOW the 3 A and 5 A fuses would melt in normal use because the current is larger than their ratings

ECF from (b): choose the lowest available rating above the answer to (b)

Worked solution

(a) Power = current × voltage, so P = I × V.

(b) First change kilowatts into watts: 1.8 kW = 1.8 × 1000 = 1800 W. Rearrange to make current the subject: I = P / V = 1800 / 230 = 7.826... A, which is 7.8 A to 2 significant figures. Check: 7.8 × 230 = 1794 W, which is about 1.8 kW.

(c) A fuse should melt only when the current is larger than it should be. Its rating must be just above the normal current. The 3 A and 5 A fuses are below 7.8 A, so they would melt every time the hairdryer was switched on. The 13 A fuse is the lowest rating above 7.8 A.

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12 · Part 1 Paper 1 & 2 Mains electricity [6]

Mark scheme

(a) **[2]**

- electrons (in the current) collide with the atoms OR ions in the resistor (1)
- energy is transferred to the atoms OR ions, so they vibrate more and the temperature increases (1)

ALLOW electrical work is done on the resistor, so the thermal energy of the element increases

(b) **[3]**

- converts time: 2.0 hours = 7200 s (1)
- substitution: E = 0.40 × 230 × 7200 (1)
- 6.6×10^5 J (1)

ALLOW 662 000 J OR 662 400 J

ALLOW 184 J OR 180 J for one mark only (time left in hours)

(c) **[1]**

- any one correct appliance, e.g. kettle, electric cooker OR oven, toaster, iron, hairdryer, electric fire OR room heater, immersion heater, electric shower (1)

Worked solution

(a) The current is a flow of electrons through the resistor. The electrons keep colliding with the atoms (ions) of the metal. Each collision transfers energy to the atoms, which vibrate more. More vibration means a higher temperature, so the element gets hot.

(b) The equation needs the time in seconds: 2.0 hours = 2.0 × 60 × 60 = 7200 s. E = I × V × t = 0.40 × 230 × 7200 = 662 400 J, which is 6.6×10^5 J to 2 significant figures. Check: the power is 0.40 × 230 = 92 W, meaning 92 J every second, and 92 × 7200 = 662 400 J.

(c) Any appliance whose job is to heat something, such as a kettle or a toaster.

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13 · Part 1

Paper 1 & 2 Mains electricity [6]

Mark scheme

(a) **[4]**

Indicative points, any four (1 each):

- the earth wire connects the metal case to the earth (1)
- the earth wire has a low resistance (1)
- a large current flows from the live wire through the case and the earth wire (1)
- this large current makes the fuse melt (1)
- the fuse is in the live wire, so melting it breaks the circuit and disconnects the case from the live supply (1)
- the case is no longer live, so a person touching it does not get an electric shock (1)

ALLOW the current flows through the earth wire rather than through the person

REJECT the earth wire "absorbs" the current OR the fuse "stops" the current without melting

(b) **[2]**

- the outer case is made of an insulator OR plastic, which does not conduct (1)
- so the case cannot become live even if a live wire touches it, so the user cannot get a shock from it (1)

ALLOW all live parts are covered by two layers of insulation, so the user cannot touch a live part

Worked solution

(a) When the live wire touches the metal case, the case becomes connected to the live supply. The earth wire gives the current a very easy path from the case to the earth, because its resistance is low. A large current therefore flows through the live wire, the case and the earth wire. This current is much larger than the fuse rating, so the fuse wire melts. The fuse is in the live wire, so once it melts the washing machine, including its case, is cut off from the live supply. Anyone who then touches the case is safe.

(b) In a double-insulated appliance the case is made of an insulator such as plastic, and the live parts are also insulated inside. Even if a wire comes loose, the case cannot carry a current to the user, so there is nothing for an earth wire to protect against.

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14 · Part 1 Paper 1 & 2 Energy and voltage in circuits [5]

Mark scheme

(a) **[1]**

- (current is) the rate of flow of charge (1)

ALLOW the charge passing a point per second

REJECT the flow of charge (no idea of rate)

(b) **[1]**

- charge = current × time (1)

ALLOW Q = I × t OR Q = It OR any correct rearrangement

(c) **[3]**

- converts time: 5.0 minutes = 300 s (1)
- substitution: Q = 0.80 × 300 (1)
- 240 C (1)

ALLOW 4.0 C for one mark only (time left in minutes)

Worked solution

(a) Current is the rate of flow of charge: how much charge passes a point in the circuit each second.

(b) Charge = current × time, Q = I × t.

(c) The time must be in seconds: 5.0 × 60 = 300 s. Q = 0.80 × 300 = 240 C. Check: 240 C ÷ 300 s = 0.80 A, which matches the current given.

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15 · Part 1 Paper 1 & 2 Energy and voltage in circuits [8]

Mark scheme

(a) **[2]**

- 9.0 J of energy is transferred (1)
- per coulomb of charge that passes (1)

ALLOW 9.0 joules per coulomb for both marks

ALLOW the energy transferred per unit charge passed is 9.0 J for both marks

ALLOW "energy per unit charge" with no number for one mark only

(b) **[1]**

- energy transferred = charge × voltage (1)

ALLOW E = Q × V OR E = QV OR any correct rearrangement

(c) **[2]**

- substitution: E = 1500 × 9.0 (1)
- 13 500 J (1)

ALLOW 1.35×10^4 J OR 1.4×10^4 J OR 14 000 J

(d) **[3]**

- rearrangement of charge = current × time: t = Q / I OR t = 1500 / 0.050 (1)
- 30 000 s (1)
- 8.3 hours (1)

ALLOW 8.33 hours OR 8 hours 20 minutes

ECF correct conversion of an incorrect time in seconds to hours

Worked solution

(a) A voltage of 9.0 V means that 9.0 J of energy is transferred for every coulomb of charge that passes. A volt is a joule per coulomb.

(b) Energy transferred = charge × voltage, E = Q × V.

(c) E = 1500 × 9.0 = 13 500 J (1.35×10^4 J). Check: 13 500 J ÷ 1500 C = 9.0 J/C = 9.0 V.

(d) Charge = current × time, so time = charge ÷ current = 1500 ÷ 0.050 = 30 000 s. To change seconds into hours, divide by 3600: 30 000 ÷ 3600 = 8.3 hours. Check: 0.050 A × 30 000 s = 1500 C.

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16 · Part 1 Paper 1 & 2 Energy and voltage in circuits [6]

Mark scheme

(a) **[1]**

- 45 Ω (1)

(b) **[1]**

- voltage = current × resistance (1)

ALLOW V = I × R OR V = IR OR any correct rearrangement

(c) **[2]**

- substitution: I = 9.0 / 45 (1)
- 0.20 A (1)

ECF from (a)

(d) **[2]**

- substitution: V = 0.20 × 30 (1)
- 6.0 V (1)

ECF from (c)

ALLOW voltage shared in proportion to resistance: (30 / 45) × 9.0 = 6.0 V

Worked solution

(a) In a series circuit the resistances add: 15 + 30 = 45 Ω.

(b) Voltage = current × resistance, V = I × R.

(c) The whole 9.0 V is across the total 45 Ω, so I = V / R = 9.0 / 45 = 0.20 A. In a series circuit this same current flows through every component, so the ammeter reads 0.20 A.

(d) For the 30 Ω resistor: V = I × R = 0.20 × 30 = 6.0 V. Check: the 15 Ω resistor has 0.20 × 15 = 3.0 V across it, and 6.0 + 3.0 = 9.0 V, the battery voltage.

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17 · Part 1 Paper 1 & 2 Energy and voltage in circuits [8]

Mark scheme

(a) **[1]**

- 7.2 V (1)

(b) **[3]**

- voltage = current × resistance OR R = V / I (1)
- substitution: R = 7.2 / 0.30 (1)
- 24 Ω (1)

ECF from (a)

ALLOW 16 Ω (resistance of the lamp) for one mark only

(c) **[2]**

- R = 12 / 0.30 OR 24 + 16 (1)
- 40 Ω (1)

ECF from (b) if the second method is used

(d) **[2]**

- the current increases (1)
- the total resistance of the circuit is smaller OR the voltage across the lamp is now larger (12 V instead of 4.8 V) (1)

REJECT the current stays the same

Worked solution

(a) In a series circuit the supply voltage is shared between the components: 12 − 4.8 = 7.2 V across the resistor.

(b) The current is the same everywhere in a series circuit, so 0.30 A flows through the resistor. R = V / I = 7.2 / 0.30 = 24 Ω.

(c) For the whole circuit, R = V / I = 12 / 0.30 = 40 Ω. Check: the lamp's resistance is 4.8 / 0.30 = 16 Ω, and 24 + 16 = 40 Ω.

(d) With the resistor removed there is less resistance in the circuit, and the lamp now has the full 12 V across it instead of 4.8 V. A larger voltage across the lamp gives a larger current, so the lamp gets brighter.

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18 · Part 1 Paper 1 & 2 Energy and voltage in circuits [7]

Mark scheme

(a) **[4]**

Indicative points, any four (1 each):

- the diode is connected in series with an ammeter and a power supply (1)
- a variable resistor OR variable power supply is used to change the voltage (1)
- a voltmeter is connected in parallel across the diode (1)
- a fixed (protective) resistor is connected in series to limit the current (1)
- change the voltage in steps and record the voltage and current each time (1)
- reverse the diode (OR the supply connections) and repeat to take readings in the other direction (1)

REJECT voltmeter in series OR ammeter in parallel for the relevant mark

(b) **[2]**

- there is no current (zero current) until the voltage reaches about 0.4 V (1)
- above this the current increases rapidly, by a larger amount for each step in voltage (1)

ALLOW current is not proportional to voltage for one mark if no other mark is scored

(c) **[1]**

- the current changes quickly (by large amounts) in this region, so more readings are needed to show the pattern (1)

ALLOW to find more precisely the voltage at which the current starts

Worked solution

(a) Connect the diode in series with an ammeter, a fixed resistor and a power supply whose voltage can be varied (or a fixed supply with a variable resistor). The fixed resistor protects the diode by stopping the current becoming too large. Connect a voltmeter in parallel across the diode only. Increase the voltage in small steps, and each time record the voltmeter and ammeter readings. Then turn the diode round and repeat to see what happens in the reverse direction.

(b) From 0.0 V to 0.4 V the current is zero: the diode does not conduct. Above about 0.5 V the current rises faster and faster: from 0.6 V to 0.7 V it goes up by 13 mA, and from 0.7 V to 0.8 V by 27 mA.

(c) Where the current changes quickly, readings that are far apart would miss the shape of the graph. Smaller steps show more clearly where and how quickly the current starts to rise.

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19 · Part 1 Paper 1 & 2 Energy and voltage in circuits [5]

Mark scheme

(a) **[4]**

Indicative points, any four (1 each):

- each lamp has the full mains voltage across it (1)
- so each lamp works at its normal brightness (1)
- each lamp can be switched on and off independently (1)
- if one lamp breaks OR blows, the other lamps stay on (1)
- adding more lamps does not make the other lamps dimmer (1)

ALLOW in series the mains voltage would be shared, so each lamp would be dim; in series one broken lamp would switch off all the lamps

(b) **[1]**

- the mains voltage is shared between the lamps, so each lamp needs only a small voltage (1)

ALLOW only one switch is needed to control all the lamps OR less wire is needed

Worked solution

(a) In a parallel circuit, components connected in parallel have the same voltage across them. So each lamp has the full mains voltage and lights at its normal brightness. Each lamp is on its own branch, so it can have its own switch, and if one lamp blows the current still flows through the other branches. In series, the voltage would be shared, so the lamps would be dim, they could not be switched separately, and one broken lamp would break the whole circuit.

(b) In series the 20 lamps share the mains voltage, so each lamp receives only a small fraction of it. Small, cheap low-voltage lamps can then be used.

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20 · Part 1 Paper 1 & 2 Energy and voltage in circuits [8]

Mark scheme

(a) **[2]**

- substitution: V = 1.5 × 6.0 (1)
- 9.0 V (1)

(b) **[1]**

- 9.0 V (1)

ECF from (a)

(c) **[1]**

- 1.0 A (1)

(d) **[2]**

- substitution: R = 9.0 / 1.0 (1)
- 9.0 Ω (1)

ECF from (b) and (c)

(e) **[2]**

- charge is not used up OR created OR stored at a junction (1)
- so the charge flowing into the junction each second equals the charge flowing out each second (1)

ALLOW the number of electrons entering the junction per second equals the number leaving per second

Worked solution

(a) Use voltage = current × resistance for the 6.0 Ω resistor: V = 1.5 × 6.0 = 9.0 V.

(b) The two resistors are in parallel, and components in parallel have the same voltage across them, so R also has 9.0 V across it.

(c) The 2.5 A from the battery splits at the junction into the two branches. The current in R is 2.5 − 1.5 = 1.0 A.

(d) R = V / I = 9.0 / 1.0 = 9.0 Ω. Check: 9.0 V ÷ 9.0 Ω = 1.0 A and 9.0 V ÷ 6.0 Ω = 1.5 A, which add to 2.5 A.

(e) Current is a flow of charge. Charge cannot pile up at a junction or disappear there, so every coulomb that flows into a junction must flow out again. The total current into a junction therefore equals the total current out of it.

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21 · Part 1 Paper 1 & 2 Energy and voltage in circuits [5]

Mark scheme

(a) **[2]**

- the current decreases (1)
- because the total resistance of the circuit increases (1)

ALLOW the battery voltage is now shared between two lamps

(b) **[2]**

- the current increases (1)
- because a larger voltage is applied across the same resistance (1)

(c) **[1]**

- zero OR no current (1)

ALLOW a very small current

Worked solution

(a) Adding a second lamp in series adds more resistance. The same 3.0 V battery now pushes charge through a larger total resistance, so the current is smaller and both lamps are dimmer than the single lamp was.

(b) The resistance of the circuit has not changed, but the voltage pushing the charge round has increased from 3.0 V to 4.5 V, so the current increases.

(c) A diode lets current flow in one direction only. Connected in reverse it has a very high resistance, so there is no current in the series circuit and the lamps go out.

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22 · Part 1 Paper 1 & 2 Energy and voltage in circuits [8]

Mark scheme

(a) **[3]**

- energy transferred = charge × voltage OR Q = E / V (1)
- substitution: Q = 3.6×10^4 / 12 (1)
- 3000 C (1)

ALLOW 3.0×10^3 C

(b) **[3]**

- converts time: 10 minutes = 600 s (1)
- substitution: I = 3000 / 600 (1)
- 5.0 A (1)

ECF from (a)

ALLOW 300 A for one mark only (time left in minutes)

(c) **[2]**

- substitution: P = 5.0 × 12 OR P = 3.6×10^4 / 600 (1)
- 60 W (1)

ECF from (b)

Worked solution

(a) Energy transferred = charge × voltage, so charge = energy ÷ voltage = 36 000 ÷ 12 = 3000 C.

(b) Charge = current × time, so current = charge ÷ time. The time must be in seconds: 10 × 60 = 600 s. I = 3000 ÷ 600 = 5.0 A.

(c) Power = current × voltage = 5.0 × 12 = 60 W. Check: power is the energy transferred per second, and 36 000 J ÷ 600 s = 60 W.

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23 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [6]

Mark scheme

(a) **[1]**

- the resistance decreases (1)

(b) **[3]**

- voltage = current × resistance OR R = V / I (1)
- substitution with current in amperes: R = 6.0 / 0.012 (1)
- 500 Ω (1)

ALLOW 0.5 Ω for two marks only (current left in mA)

(c) **[2]**

- the ammeter reading increases (1)
- because the resistance of the thermistor decreases as it warms up (1)

ECF from (a)

Worked solution

(a) As a thermistor gets warmer its resistance decreases.

(b) First change milliamperes into amperes: 12 mA = 12 ÷ 1000 = 0.012 A. Then R = V / I = 6.0 / 0.012 = 500 Ω. Check: 6.0 V ÷ 500 Ω = 0.012 A = 12 mA.

(c) When the door is open the thermistor gets warmer, so its resistance falls. The battery voltage stays the same, so with less resistance a larger current flows and the ammeter reading goes up.

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24 · Part 1 Paper 2 only Electric charge [5]

Mark scheme

(a) **[3]**

- rubbing transfers electrons between the cloth and the rod (1)
- polythene is an insulator, so electrons cannot move through it and the charge stays on the rod (1)
- copper is a conductor, so electrons flow through the rod and Mia's body to OR from the earth, so no charge builds up (1)

ALLOW copper is a conductor so the charge is conducted away through her hand

(b) **[2]**

- yes, the copper rod becomes charged (1)
- the plastic handle is an insulator, so electrons cannot flow between the rod and Mia's hand OR the earth (1)
- the reason mark is not given without a correct decision

Worked solution

(a) Rubbing two materials together moves electrons from one to the other. On the polythene rod the electrons cannot move, because polythene is an insulator, so the extra (or missing) electrons stay where they are and the rod is charged. Copper is a conductor. Electrons can move freely through the copper and through Mia's body to or from the earth, so any charge flows away as soon as it is produced.

(b) With the plastic handle between the copper and Mia's hand, there is no conducting path to the earth. Electrons transferred by rubbing cannot escape, so the copper rod stays charged.

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25 · Part 1 Paper 2 only Electric charge [7]

Mark scheme

(a) **[2]**

- negative (1)
- it repels the negatively charged hanging rod, and like charges repel (1)
- the reason mark is not given without the correct charge

(b) **[2]**

- an uncharged rod would also be attracted to the charged hanging rod (1)
- because the negative hanging rod repels electrons in the uncharged rod away from the near end, so the near end becomes positive and is attracted (1)

> **ALLOW** attraction happens between unlike charges and also between a charged and an uncharged object, so only repulsion proves the type of charge

(c) **[2]**

- hold the rubbed perspex rod near a hanging object known to be positively charged (1)
- if the two repel, the perspex rod is positively charged (1)

> **ALLOW** rub a second perspex rod in the same way and show that the two perspex rods repel each other (so the test rod is charged) (1); since the charged perspex rod is attracted to the negative hanging rod, its charge must be positive (1)

(d) **[1]**

- any one of: the same cloth OR same material of cloth; the same number of rubs; the same force OR pressure of rubbing; the same distance between the test rod and the hanging rod (1)

Worked solution

(a) The polythene test rod pushes the hanging rod away. Repulsion happens only between like charges, and the hanging rod is negative, so the test rod is negative too.

(b) Attraction happens between unlike charges, but a charged object also attracts an uncharged one. The negative hanging rod repels electrons in a nearby uncharged object, so the near side of that object becomes slightly positive and is pulled towards it. The perspex rod could therefore be positive or uncharged.

(c) Repulsion is the reliable test. If the perspex rod is held near something that is known to be positively charged and the two push apart, the perspex must be positive.

(d) To make the test fair, the way the rod is charged must be the same each time, for example using the same cloth and the same number of rubs.

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26 · Part 1 Paper 2 only Electric charge [6]

Mark scheme

(a) **[2]**

- rubbing (friction) transfers electrons (1)
- from the hair to the balloon, so the balloon gains electrons (1)

REJECT positive charges OR protons move, for the second mark

(b) **[2]**

- positive (1)
- the hair has lost electrons (1)
- the explanation mark is not given without the correct charge

(c) **[2]**

- all the hairs have the same type of charge OR all the hairs are positively charged (1)
- like charges repel (1)

Worked solution

(a) When the balloon is rubbed on the hair, friction moves electrons from the hair onto the balloon. Electrons are negative, so the balloon now has more electrons than protons and is negatively charged.

(b) The hair has lost those electrons, so it has more protons than electrons and is positively charged. Only electrons move; the positive charges stay where they are.

(c) Every hair has lost electrons, so every hair is positively charged. Like charges repel, so the hairs push each other apart.

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27 · Part 1 Paper 2 only Electric charge [6]

Mark scheme

(a) **[3]**

- friction between the fuel and the hose transfers electrons (1)
- so charge builds up on the aircraft OR the fuel OR the hose (1)
- a spark may jump, which could ignite the fuel (vapour) and cause a fire OR explosion (1)

(b) **[2]**

- the metal cable is a conductor, so electrons flow between the aircraft and the earth (1)
- so charge cannot build up and no spark is produced (1)

ALLOW the aircraft is earthed, so the charge flows away safely

(c) **[1]**

- any one sensible situation, e.g. refuelling a fuel tanker OR road tanker OR ship; filling a car at a petrol station; powder OR grain OR flour flowing through pipes or into silos; working near flammable gases (1)

ALLOW lightning OR charge building up on clouds

Worked solution

(a) As the fuel rushes along the hose, friction between the fuel and the hose moves electrons from one to the other. Charge builds up on the aircraft and the fuel. If enough charge builds up, a spark can jump between the aircraft and the nozzle or the ground. Fuel vapour is very flammable, so a spark could start a fire or an explosion.

(b) The metal cable is a conductor joining the aircraft to the earth. Electrons can flow along it, so any charge produced flows away to the earth as soon as it forms. The charge never becomes large enough to make a spark.

(c) Any situation where friction can build up charge near something that burns easily, for example filling a road tanker with fuel.

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28 · Part 1 Paper 2 only Electric charge [5]

Mark scheme

(a) **[2]**

- upwards OR towards the upper (positive) plate (1)
- because unlike charges attract: the negative drops are attracted to the positive plate (and repelled by the negative plate) (1)
- the explanation mark is not given without the correct direction

(b) **[2]**

- more charge on the plates gives a larger (electrostatic) force on each drop (1)
- so the drop is deflected more OR lands further up the paper (1)

(c) **[1]**

- it is not deflected by the plates OR it continues in a straight line (1)

ALLOW it is not attracted to either plate

Worked solution

(a) The drops are negative. The upper plate is positive, so it attracts them, and the lower plate is negative, so it repels them. Both forces push the drops upwards.

(b) A larger charge on the plates means a stronger force on each charged drop. The drop is pushed further upwards while it is between the plates, so it lands at a different, higher point on the paper. By changing the charge, the printer can place each drop exactly where it is needed.

(c) An uncharged drop has no charge for the plates to attract or repel, so it carries straight on without being deflected by the plates.

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29 · Part 1 Paper 2 only Electric charge [6]

Mark scheme

(a) **[5]**

Indicative points, any five (1 each):

- a drum OR plate is given an electrostatic charge (1)
- an image of the page is formed on the drum using light reflected from the page (1)
- where light falls on the drum (the white parts of the page) the charge leaks away OR is lost (1)
- the dark parts (the printing) stay charged, leaving a pattern of charge in the shape of the image (1)
- toner powder with the opposite charge is attracted to the charged areas of the drum (1)
- a sheet of paper is charged (more strongly) so that the toner is attracted from the drum onto the paper (1)
- the paper is heated so that the toner melts and sticks to it (1)

ALLOW any consistent choice of signs, as long as the toner and the charged drum areas have opposite charges

(b) **[1]**

- the particles repel each other so they spread out evenly OR do not clump together (1)

ALLOW so that they are attracted only to the oppositely charged parts of the drum

Worked solution

(a) The drum inside the photocopier is first given a charge all over. Light is shone onto the page, and the reflected light forms an image of the page on the drum. Where light hits the drum (the white parts of the page), the drum conducts and its charge leaks away. Where the page is dark (the writing), no light arrives, so the drum stays charged. This leaves a pattern of charge on the drum that matches the printing. Toner powder with the opposite charge is attracted to these charged areas. Paper is then given a charge that pulls the toner off the drum onto the paper, and the paper is heated so the toner melts and sticks.

(b) Like charges repel, so toner particles with the same charge push each other apart. They spread out evenly instead of forming clumps, and they stick only to the oppositely charged areas of the drum.

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30 · Part 1 Paper 2 only Electric charge [6]

Mark scheme

(a) **[4]**

Indicative points, any four (1 each):

- friction between the shoes and the carpet transfers electrons (1)
- so Ana becomes charged (1)
- the rubber soles are an insulator, so the charge cannot flow away to the earth and it builds up (1)
- the metal door handle is a conductor (connected to the earth) (1)
- when she touches it, electrons flow quickly between Ana and the handle OR earth (1)
- this flow of charge through her body is a current, which she feels as a shock OR spark (1)

(b) **[2]**

- substitution: $I = (3.0 \times 10^{-7}) / (1.5 \times 10^{-6})$ (1)
- 0.20 A (1)

ALLOW 0.2 A

Worked solution

(a) As Ana walks, her rubber soles rub on the nylon carpet and electrons are moved between them, so she becomes charged. Rubber is an insulator, so the charge cannot escape through her shoes to the ground, and it builds up on her body. The metal handle is a conductor connected to the earth. When she touches it, the built-up charge suddenly flows through her hand to the handle (or from it). This sudden flow of charge is an electric current, and she feels it as a shock.

(b) Charge = current × time, so current = charge ÷ time = $(3.0 \times 10^{-7}) \div (1.5 \times 10^{-6}) = 0.20$ A. Check: $0.20 \text{ A} \times 1.5 \times 10^{-6} \text{ s} = 3.0 \times 10^{-7}$ C.

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